TITILE：An Explainable AI-Integrated Chatbot System for Intelligent Health Insurance Claim Verification and Fraud Detection

ABSTRACT :

The proposed AI project aims to develop a system for verifying claims in the health insurance industry, enhancing fraud detection with improved accuracy, transparency, and efficiency. This proposed system will consist of SVM (Support Vector Machine) and XGBoost classifiers for detecting fraud in the claims. The OCR technology will be used for extracting text from the uploaded documents and further processing for predicting fraud. In order to mitigate the problem with the class imbalance in this case, SMOTE combined with random under-sampling will be used in the model.

Moreover, the proposed system will include an AI-based chatbot that can assist the user during the claim process. For this reason, SHAP (Shapley Additive Explanations) will be used to explain the main features important to the specific prediction. The project will be implemented as a secure web-based application that will allow uploading documents and receiving information on the fraud probability score.

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